

Drought /Water Reservoirs link to Mycotoxins in Kongwa

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Discussion

Levels of mycotoxins in the food stream is a possible indication of plant stress due to drought conditions. We look at a number of approaches to investigate this relationship in Kongwa, as well as neighbouring districts To account for possible external sources of food, possibly food markets in neighbouring districts.

SPEI, one of the most sophisticated approaches to assessing drought conditions, provides a monthly estimation of the water availability in the landscape. It is a key indicator of drought as it provides a clear understanding how much water is retained near the surface as a function of temperature, precipitation and evapotranspiration. It can be calculated at different timescales, for example;

- Short (3-month): Tracks recent conditions and is mainly an indicator of water availability as dominated by prior 3 months of precipitation.(**FIG 1**)
- Long (48-months) - Probe water accumulations which also affect droughts.(**FIG 2**)

A time series for long-timescale SPEI data for Kongwa alerts us to when the conditions in Kongwa shifted, roughly before 2018, Kongwa was in drought which likely relates to the high levels of mycotoxins reported in the food stream during food sampling in 2015. And just before the MMT trial, we see a transition to abundance. This strongly suggests plant stress was lower and resulted in the low levels of mycotoxins recorded from the trial. Time series in expanded concentric squares outside of Kongwa, show not much variation with the drought to abundance trends suggesting that food grown outside of Kongwa would potentially contain similar levels of mycotoxins

GRACE is an alternate method of probing water accumulation. TWSA is not an absolute measurement of the current water availability in the landscape but rather a relative measure of that particular month's value to the baseline (average of calendar months (2004-2009.9) and sensitive to water both above and below the surface. A surge in Kongwa just before the MMT trial was also seen in the GRACE data. Like the SPEI data, similar changes in TWS were also present in the surrounding districts

DSI, a computation derived from the GRACE data, also supports the SPEI data with identical transitions to abundance in 2018. DSI is an indicator of how dry or wet the landscape is in comparison to the average of similar calendar months, however it takes into account the accumulation from groundwater.

We determined that possible underground water does not play a substantial part in Kongwa. According to a geological study attempting to map groundwater potential sites in Kongwa, it was determined that the groundwater yields to an approximate modest flow of 3.5 litres per second (also backed by borehole yields).(Zacharia et al, 2025).We therefore conclude that SPEI data is more compatible for investigating the climatic conditions in Kongwa

Resorting to elevation data to determine the surface accumulation of the region, we note that Kongwa lies on a higher elevation than any nearby surface reservoirs and therefore conclude that, there is no physical possibility that the Mtera and Sulunga reservoirs were sustaining the region during this period. This finally points to soil moisture as our plausible water reservoir for longer term water storage. Supporting observations detail Kongwa to contain mostly sandy clay loam soils and sandy loam soils (Nhudha et al, 2024). They are suitable for maize production since they are capable of holding water for relatively longer periods (Kyveryga et al., 2009). This likely explains why there was a drop in the mycotoxin levels during the MMT, as crops could have possibly experienced less water-stress from the abundance of water accumulation in the soil.

Overall, SPEI, GRACE and DSI present a distinctive view of the water accumulation of water in Kongwa due to increased precipitation right before the MMT and elevation has ruled out the possibility of surface accumulation in Kongwa, leaving soil moisture (backed by Kongwa's soil types) as the viable conclusion for the water reservoir in Kongwa that caused the drought to end, with water being stored in the soil and released with a lag.

Directions For Future Research

Particular to Africa with several semi-arid regions with little to no ground monitoring, remote sensing with the same methodology (SPEI, GRACE (using downscaled- Gou & Soja data for better resolution), DSI) and taking into account human activity and climatic conditions, could be applied to determine where in the landscape agricultural crops are likely at higher risk of exposure to mycotoxins.

References

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Abstract

A health trial in between 2018 and 2020 which had documented the presence of mycotoxins (toxins made by endemic fungi) in food was funded, but ended inconclusively when mycotoxins disappeared during the trial itself. In 2024, a climatic follow-up determined that a long-term drought seemed to have ended between 2015 and the trial . This study describes total water storage (TWS) from 2002 to 2022 using remote sensing from high resolution Gravity Recovery and Climate Experiment (GRACE) data (Gou & Soja 2024), computed Drought Severity Index (DSI), and Standard Precipitation Evapotranspiration index (SPEI), topography, to describe and characterize the area the trial was carried out.

Introduction

- **The Human impact:** Childhood stunting carries life long effects including increased mortality from infectious diseases and delayed schooling. One of the potential causes is mycotoxins, especially aflatoxins: *Aspergillus* invades drought-stressed crops, so toxin risk rises with crop water stress (Phillips et al, 2024).
- Like many sub-Saharan regions in Africa, Kongwa was confirmed as a region with high mycotoxin levels in the food stream through preliminary food sampling done by Henry Wells and colleagues in 2015.
- A Gates Foundation funded Mycotoxin Mitigation Trial (MMT) starting in 2018 attempting to reduce mycotoxin levels and study the effect on infants. Identified households in the intervention group were provided with mycotoxin-free flour monthly, and periodic Urinary and blood analysis was then carried out to determine whether this affected the levels of mycotoxins in the bloodstreams of infants. Results from the control and intervention groups were statistically indistinguishable.
- **What does this imply?** : Follow up in 2024 (Liddell & Hackman) discovered long-timescale SPEI indicating that a drought ended prior to the start of the trial, which rainfall itself could not account for.
- **This current project:** We assume water was retained in the landscape. What natural reservoir-soil moisture, shallow groundwater or seasonal wetland maintained and ended that drought in the rain-fed districts with almost no irrigation or constructed storage.

Materials and Methods

Several indications of water availability at different depths and timescales were analysed throughout the course of this project.

- **SPEI** - an important drought index with standardized precipitation minus evapotranspiration (particularly a crop-relevant surface indicator). It is an indicator that is directly tied to plant water stress and therefore mycotoxin risk. We extracted data at 3-, 12-, 48- month timescales over Kongwa and used concentric rings to compare local vs regional trends.
- **GRACE-SeDA** - Provided an estimate of the TWS of Kongwa (both surface and below ground), which is determined by measuring the gravity shifts as a result of mass (water) accumulation.(Gou & Soja 2024, Nature Water)
- **DSI** - Following an established methodology, we convert the monthly Total Water Storage Anomalies (TWSA) to a drought index: standardize each calendar month anomaly, (value – month-mean)/SD, and bin into 11 drought/wet classes.
- **Terra Climate:** A dataset that provided monthly soil moisture, maximum and minimum temperatures, and precipitation.
- **Elevation:** Topographic analysis using Google Earth Engine (GEE) to analyse the terrain of the Kongwa landscape. GEE was also used to map out concentric squares highlighting regions outside of Kongwa, to address the possibility that the locals fed on maize and groundnuts that was grown outside of Kongwa.
- **Software:** GEE (Javascript) + Google Colab (Python- matplotlib, pandas, numpy, xarray) + Excel.

Results

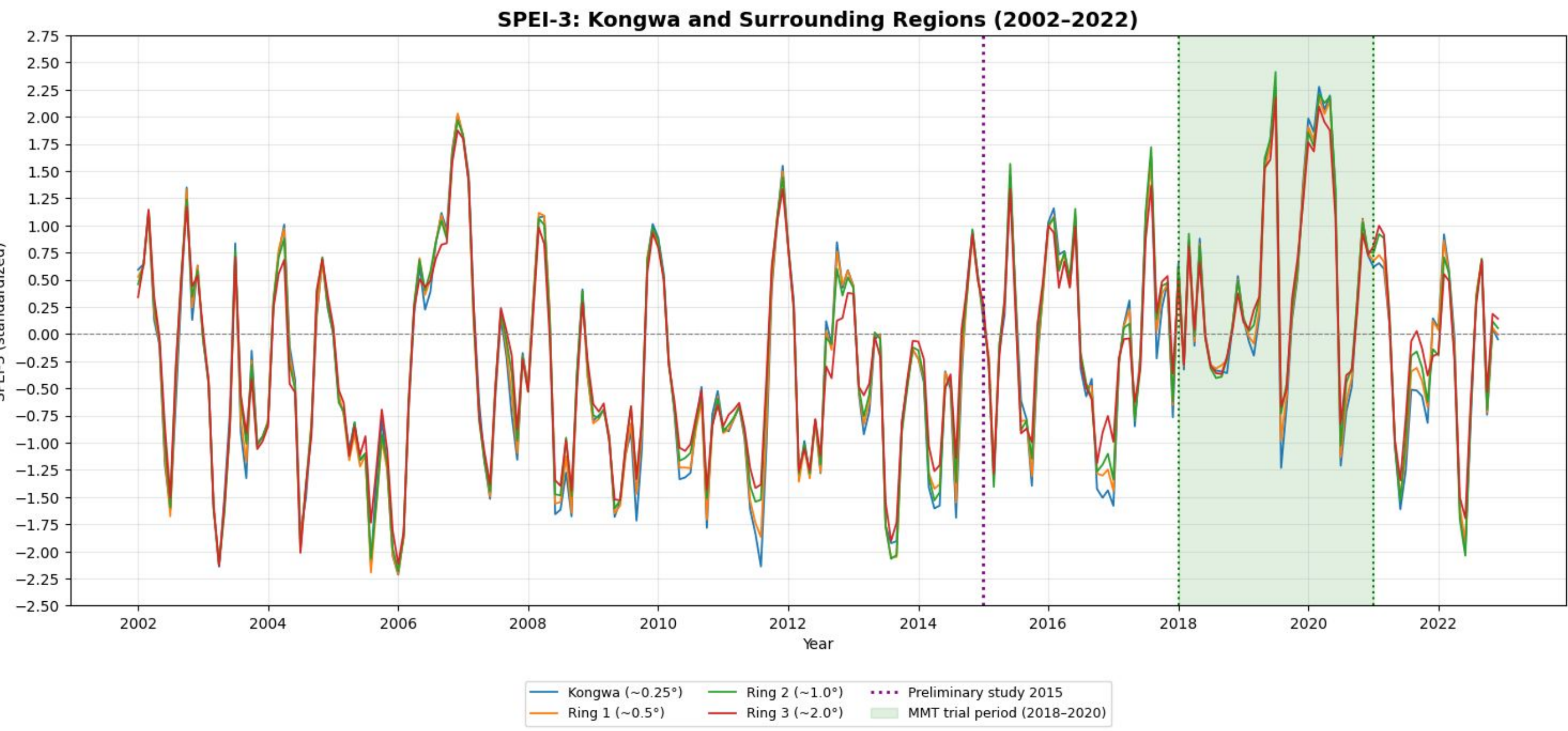


Figure 1: The figure shows the 3-month SPEI time series for Kongwa (blue line). The corresponding time series for the surrounding concentric regions (Rings 1, 2, and 3) are also displayed in different colours. The baseline value (0) indicates the normal (average) TWSA of a particular month and, different points on the graph indicate how monthly TWS varies accordingly.(negative values: dry, positive: wet).

- Evident from the graph is a high correlation between the SPEI data of Kongwa and its surrounding regions, indicating the water accumulation and loss to evapotranspiration is somewhat similar.

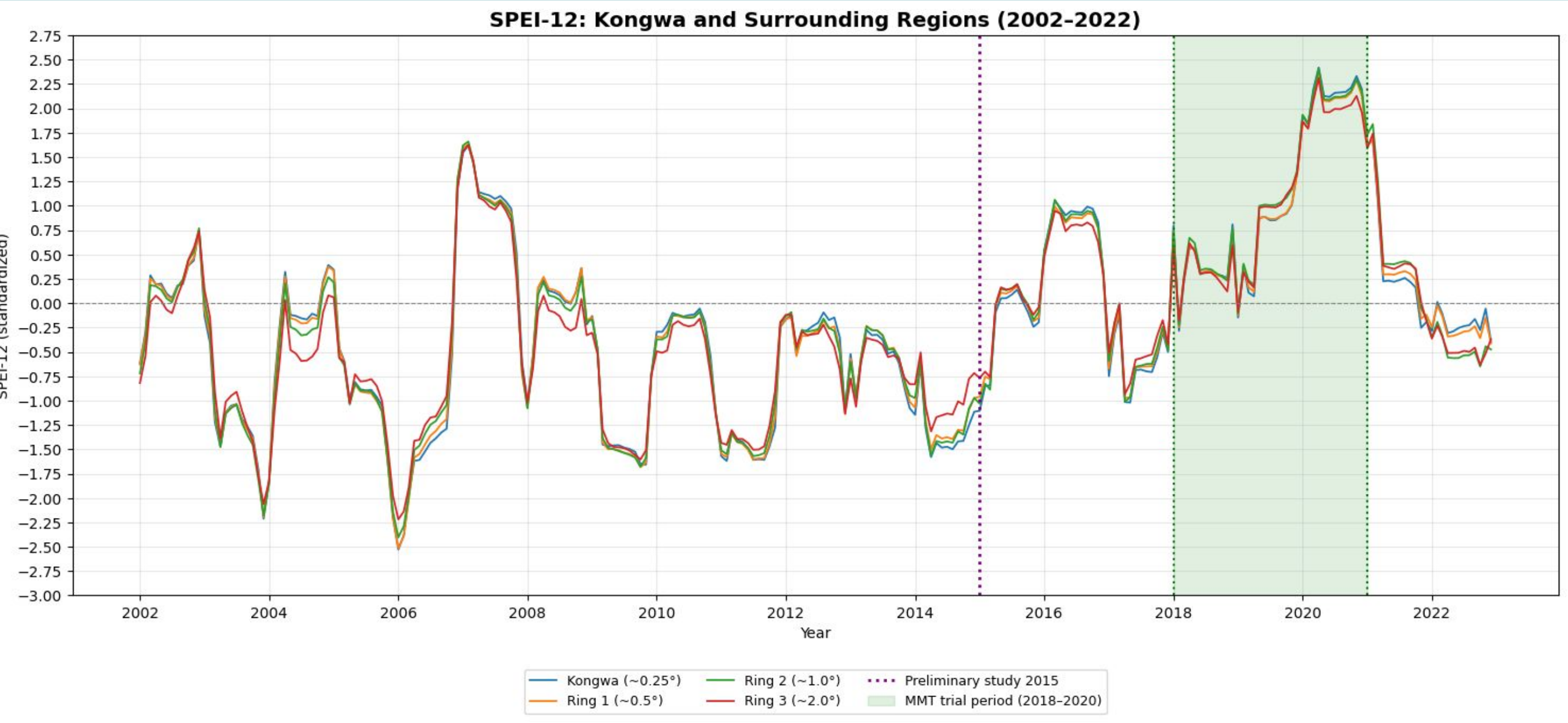


Figure 2: 48-month SPEI time series for Kongwa (blue line), alongside the time series of the surrounding concentric regions (Rings 1, 2, and 3), each represented by a different colour. As supporting evidence, the graph shows a sudden increase in the SPEI values, right before 2018 (start of MMT trial). This shows that Kongwa and surrounding regions experienced a drastic shift from drought to abundance.

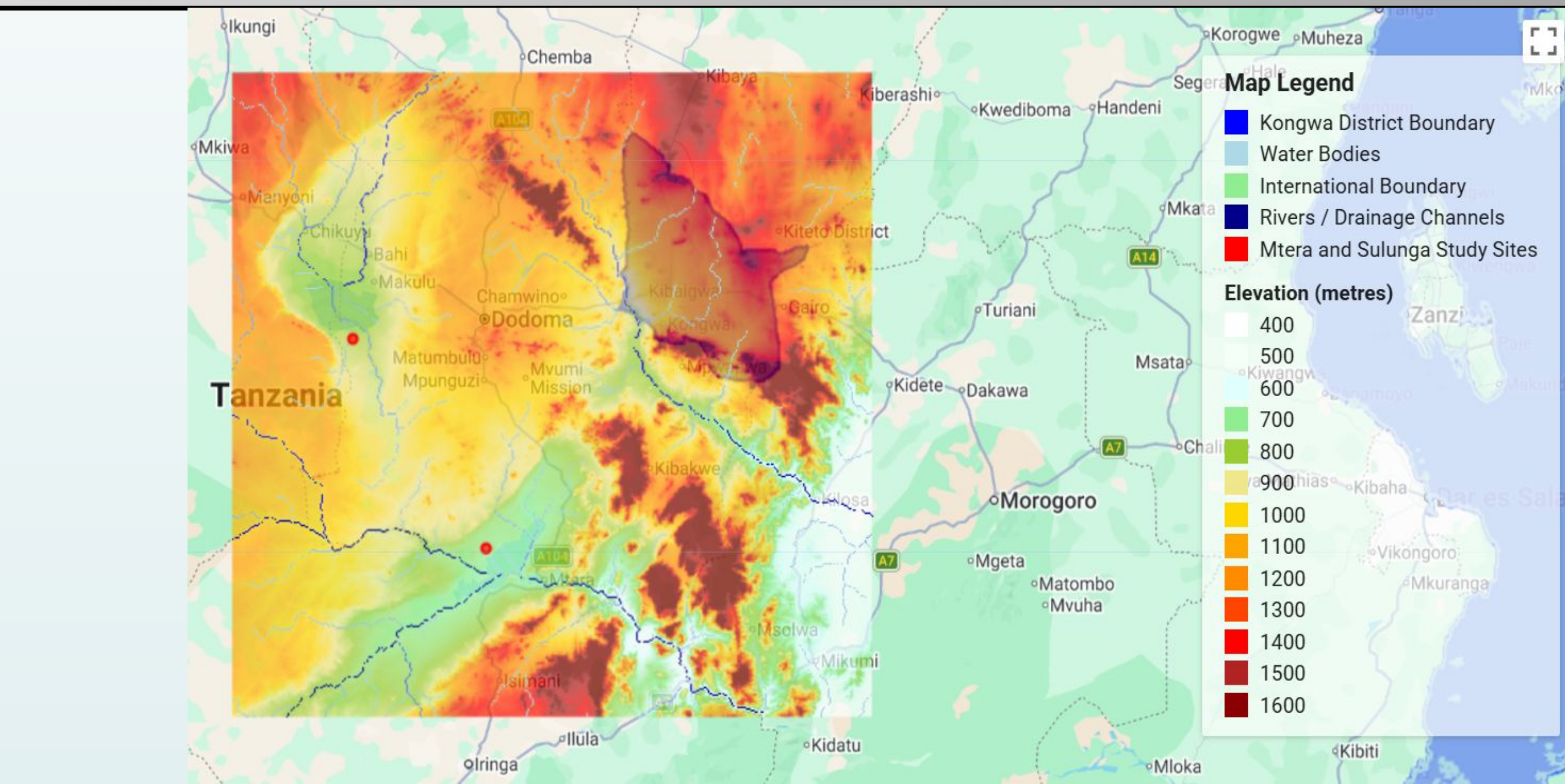


Figure 3 (left): Elevation of the Dodoma Region, with emphasis on Kongwa District.. Located outside of Kongwa, the two red dots mark the nearby major water bodies: Mtera Reservoir (lower), and Lake Sulunga(upper). These two reservoirs are situated at lower elevations relative to Kongwa

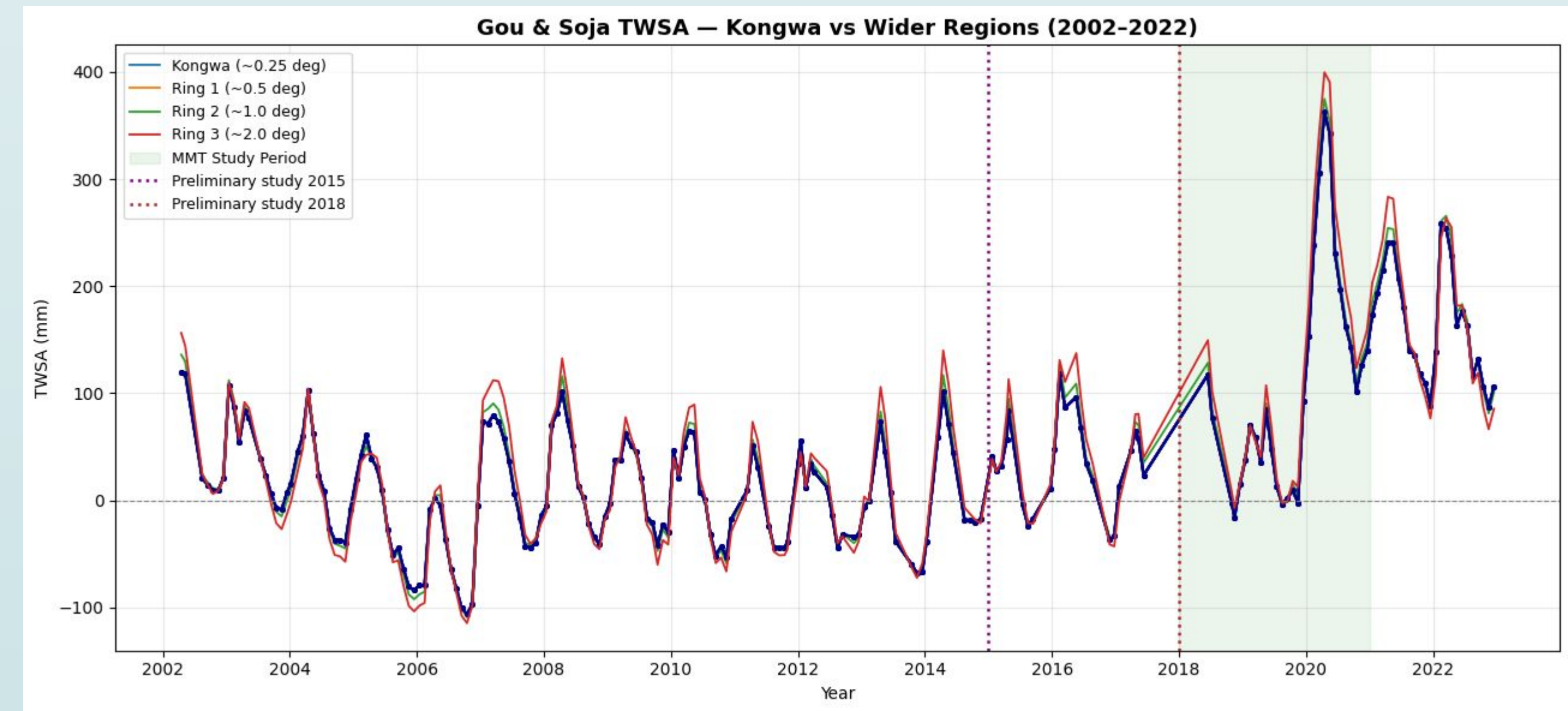


Figure 4: Shows the time series for the TWSA data according to GRACE-SeDA from 2002-2022. The graph shows a rise and fall in the TWSA of Kongwa between 2002-2018, indicating a normal annual variations between the dry and wet seasons in the region. However, after 2020, a sudden spike is noted indicating a change from drought to abundance of water in the region.

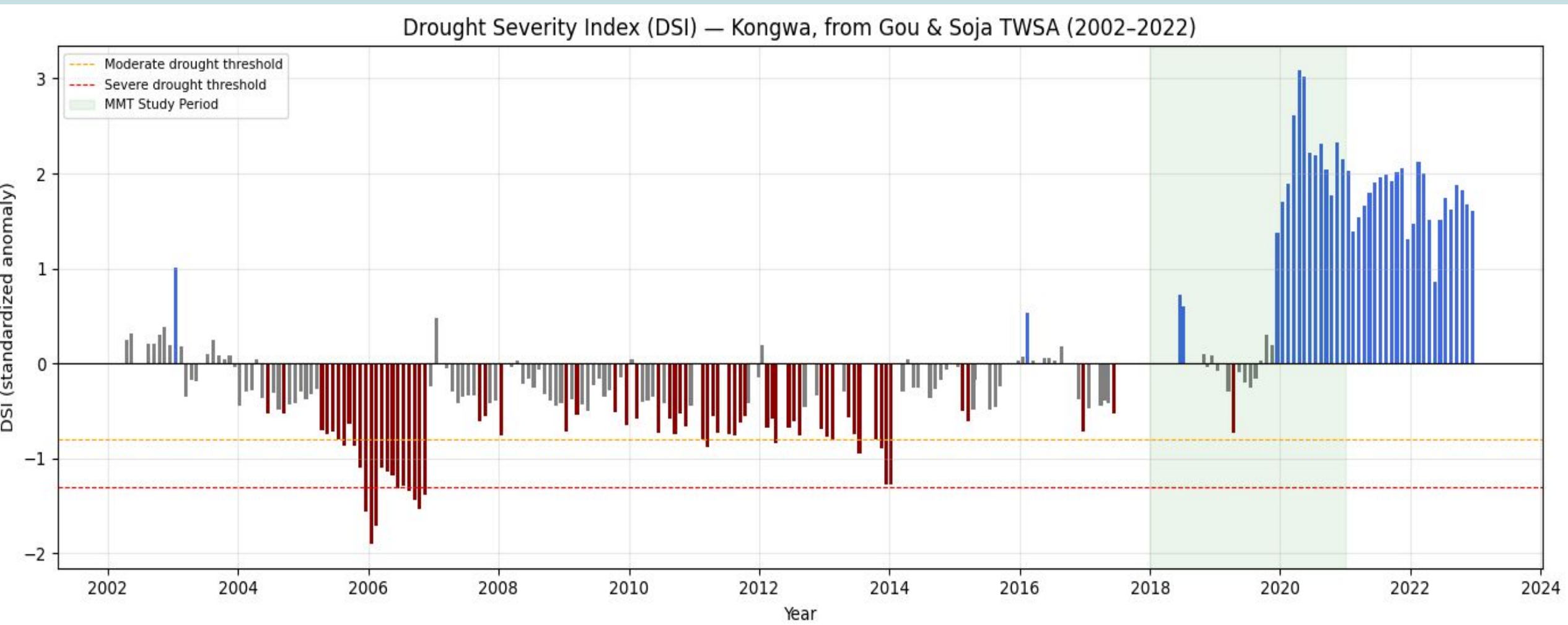


Figure 5: DSI of Kongwa, this is also another indicator in of the sudden shift from drought to abundance conditions right before 2018 and during the MMT period.

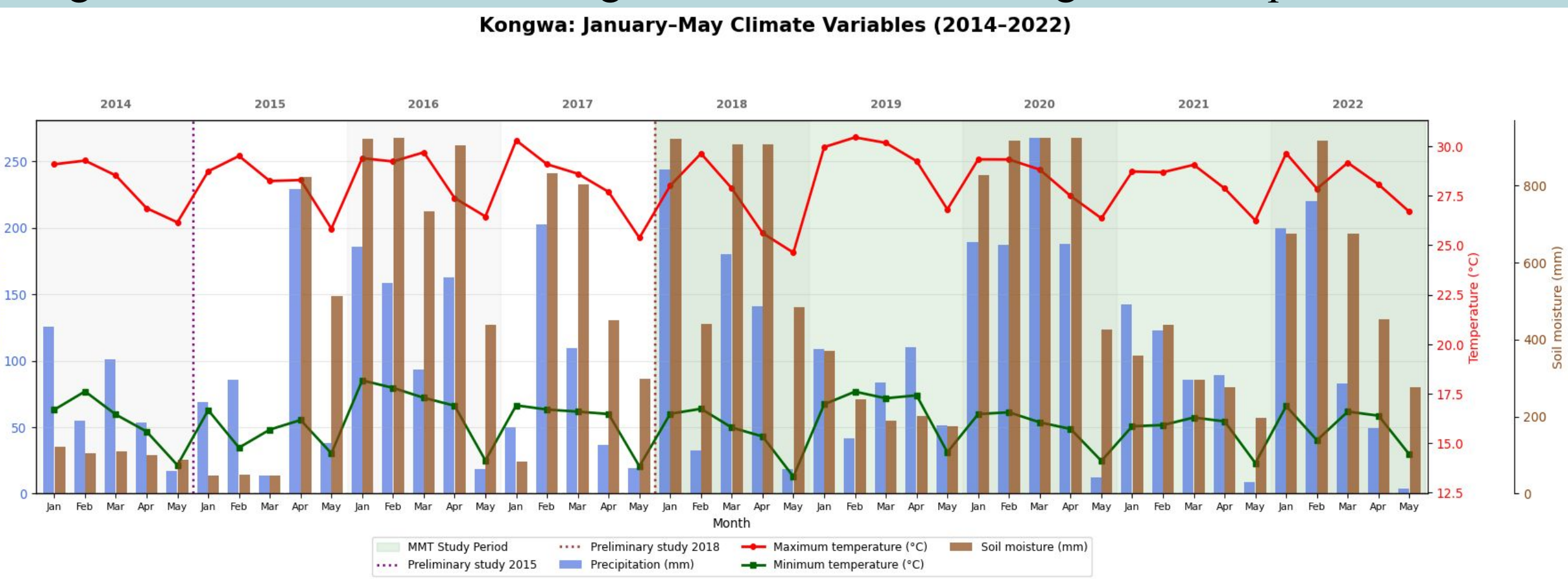
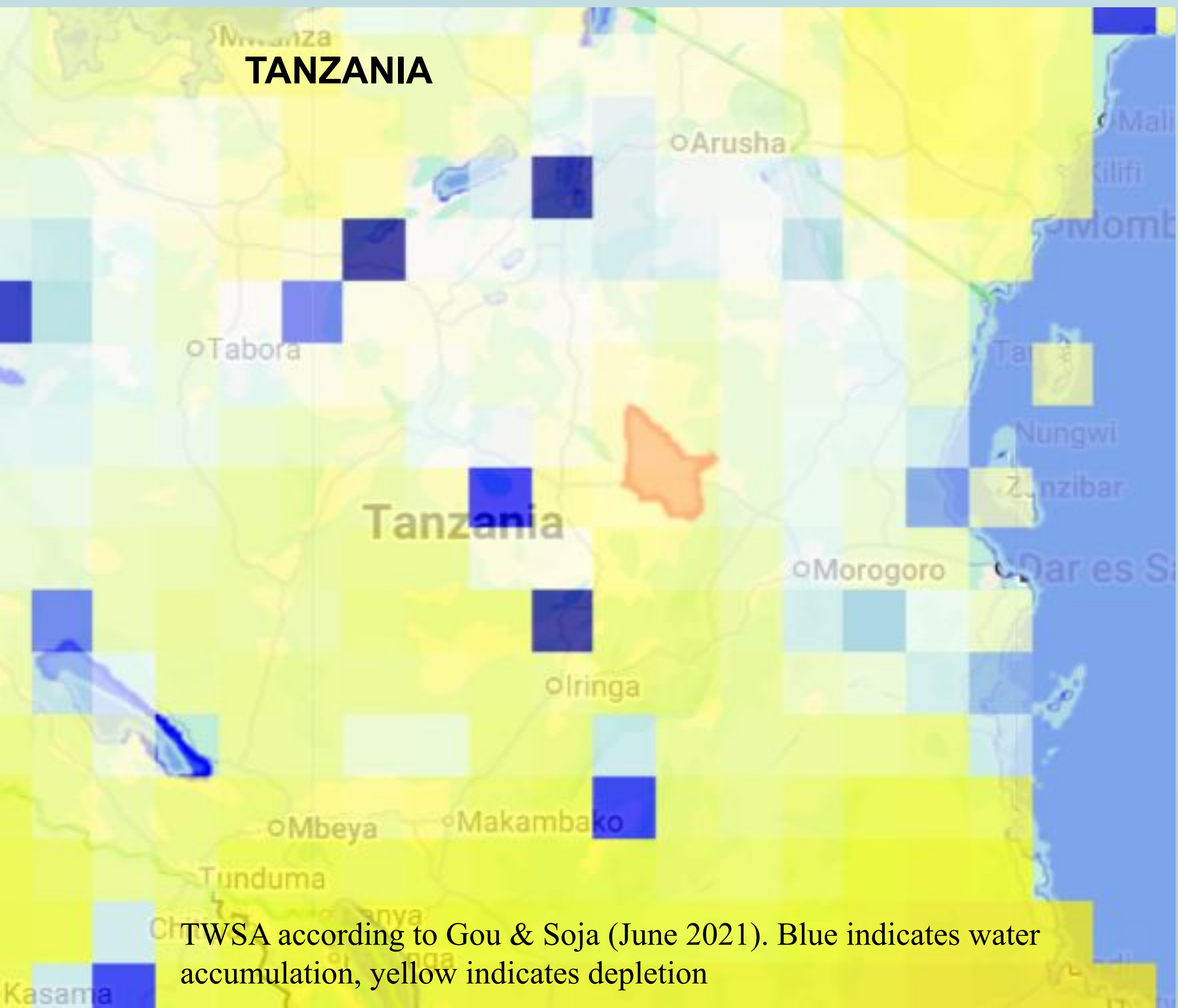


Figure 6: monthly climatic variables; maximum and minimum temperature (°C), cumulative soil moisture (mm) and precipitation (mm) in the region of Kongwa for the first half of each year from 2014 through 2022.



TWSA according to Gou & Soja (June 2021). Blue indicates water accumulation, yellow indicates depletion